

Pharmacology of Antihypertensive Agents:

- **Alpha Blockers**
- **Centrally Acting Sympatholytics**
- **Vasodilators**

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**I am available to groups and individuals for
pharmacology help and discussions by appointment.**

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Following this session, students will be able to:

1. Interpret the pharmacokinetic properties of the antihypertensive agents in the context of their clinical significance.
2. Relate the mechanisms of action of the alpha blockers, centrally acting sympatholytics, vasodilators, ganglion blocking agents and adrenergic neuron blocking agents to their cardiovascular effects by giving examples of therapeutic applications.
3. Predict the adverse effects, drug interactions, and contraindications when given the mechanism of action and/or pharmacokinetic properties of the individual drug classes.
4. Select the appropriate antihypertensive agent and monitoring parameters for the individual patient based on the drug's pharmacologic effects when presented with a clinical case vignette.

Preparation Materials (links are in the CPG and on the next slide)

Required

- ScholarRx Bricks | Practice Questions and Clinical Vignettes

Highly relevant optional materials:

- Dr. Goldstein's Word handout | Video Lecture | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them.*
- *Work through the GUIDED READING QUESTIONS with pen/pencil and paper.*

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- *Practice questions (not graded): Simple Recall and Case Vignettes*

Links to resources

Scholar Rx Brick (Required)

Antihypertensive drugs <https://exchange.scholarrx.com/brick/antihypertensive-drugs>

Renin-Angiotensin-Aldosterone System Inhibitors <https://exchange.scholarrx.com/brick/renin-angiotensin-aldosterone-system-inhibitors>

Helpful bricks on blood pressure physiology (Recommended but not required):

Autonomic regulation of the cardiovascular system <https://exchange.scholarrx.com/brick/autonomic-regulation-of-the-cardiovascular-system>

Hypertension: Foundations and Frameworks <https://exchange.scholarrx.com/brick/hypertension-foundations-and-frameworks>

Primary (Essential) Hypertension <https://exchange.scholarrx.com/brick/primary-essential-hypertension>

Baroreceptor regulation of blood pressure <https://exchange.scholarrx.com/brick/baroreceptor-regulation-of-blood-pressure>

Humoral regulation of blood pressure <https://exchange.scholarrx.com/brick/humoral-regulation-of-blood-pressure>

Textbooks/Exam review books you might find helpful:

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024

Chapter 11: Antihypertensive Agents

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281748109>

Chapter 17: Vasoactive Peptides > Angiotensin

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281749431>

LWW Health Libraries Premium Basic Sciences Lippincott Illustrated Reviews: Pharmacology, 8e, 2022; Chapter 8: Antihypertensives

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253325981&bookid=3222>

Access Medicine Katzung's Pharmacology: Examination and Board Review, 14e, 2024; Chapter 11: Drugs Used in Hypertension

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285590661>

- For tips on learning skills based on the science of learning and for “What you need to know”, please see Lecture #35, slides 6-9.

ALPHA BLOCKERS

(α adrenergic receptor blockers)

Selective alpha-1
receptor blockers

- * Prazosin
- * Terazosin
- * Doxazosin

Nonselective
alpha blockers

- * Phenoxybenzamine
- * Phentolamine

Comparing the properties of the alpha blockers

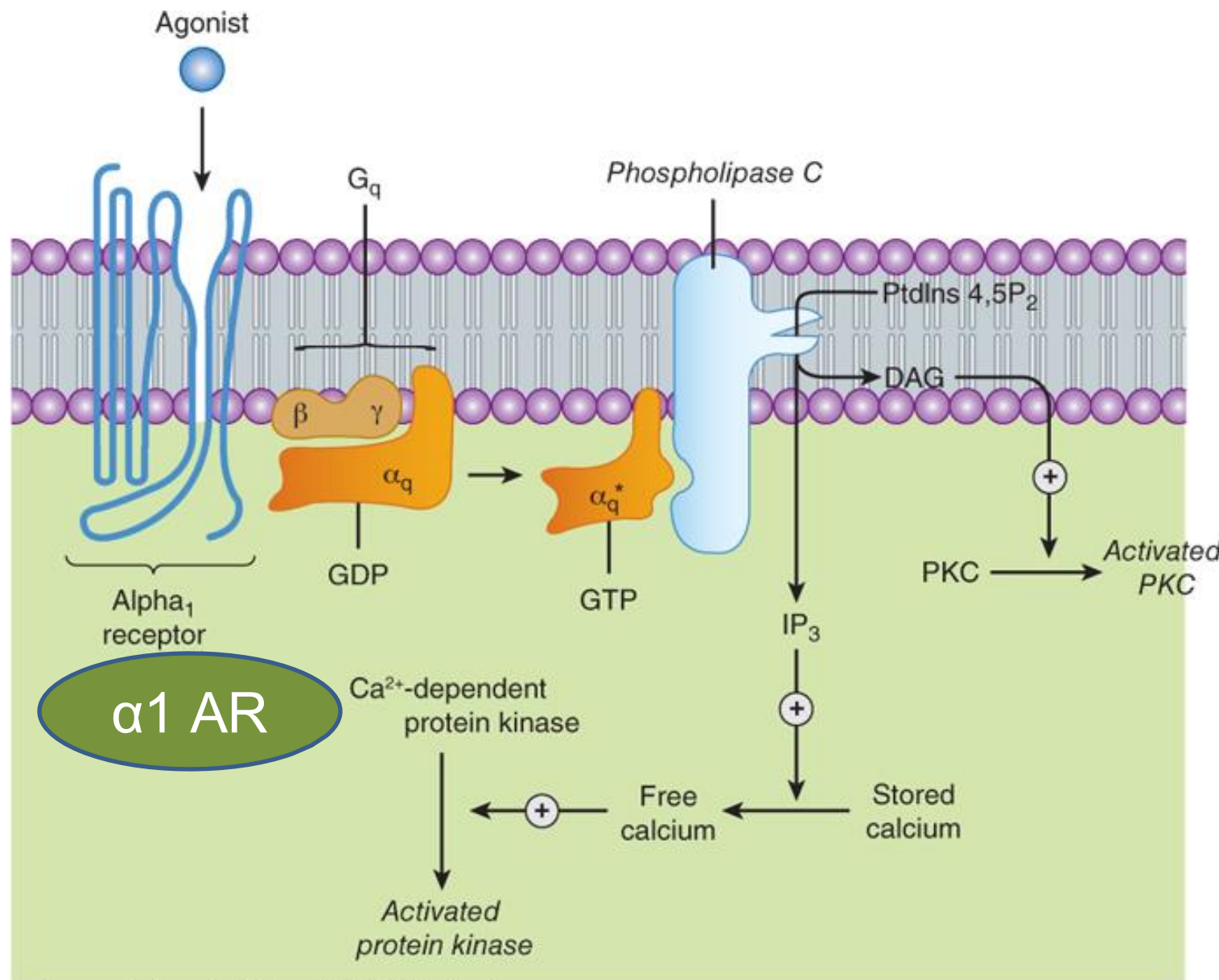
	DRUG	RECEPTOR AFFINITY	METABOLISM	ORAL AVAILABILITY (%)	PLASMA $t_{1/2}$	DURATION and DOSING
selective agents	Prazosin	α_1 selective	hepatic	60-80	2-3h	dosing 2-3x /d
	Terazosin	α_1 selective	hepatic	~100	~12h	duration ~24h 1x/d
	Doxazosin	α_1 selective	hepatic CYP3A4 (2D6, 2C19)	~65%	~22h	duration >24h dosing 1x/d
nonselective agents	Phenoxybenzamine	$\alpha_1 > \alpha_2$	not known	20-30	24h (oral)	duration ≥ 3 days
	Phentolamine	$\alpha_1 = \alpha_2$	hepatic	N/A	19 min (IV)	duration 10-30 min (IV)

α_1 -adrenergic receptors are G_q PCRs.

Agonist activation of G_q -coupling protein activates PLC, which leads to

(1) release of IP_3 , which increases intracellular Ca^{2+} , and

(2) release of DAG, which activates PKC.



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Activation of α_1 responses. Stimulation of α_1 receptors by catecholamines leads to the activation of a G_q -coupling protein. The activated α subunit (α_q^*) of this G protein activates the effector, phospholipase C, which leads to the release of IP₃ (inositol 1,4,5-trisphosphate) and DAG (diacylglycerol) from phosphatidylinositol 4,5-bisphosphate (PtdIns 4,5P₂). IP₃ stimulates the release of sequestered stores of calcium, leading to an increased concentration of cytoplasmic Ca²⁺. Ca²⁺ may then activate Ca²⁺-dependent protein kinases, which in turn phosphorylate their substrates. DAG activates protein kinase C (PKC). GDP, guanosine diphosphate; GTP, guanosine triphosphate. See text for additional effects of α_1 -receptor activation.

α_1 ARs mediate smooth muscle contraction and promote vascular smooth muscle cell and cardiac cell growth.

Effects of Stimulating Vascular Adrenergic Receptors

- α_1 receptors are the predominant subtype in most arteries and veins.
- Vasoconstriction is the predominant effect.

SNS review of vascular actions for context:

- The β_2 receptor response (vasodilation) predominates in blood vessels of skeletal muscle.
- α receptors are also present in these vascular beds.
- The renal and mesenteric vessels also contain specific dopaminergic receptors, which cause dilation when activated.

SNS: sympathetic nervous system

Effects of BLOCKING alpha-1 receptors on vascular smooth muscle.

MECHANISM	EFFECT
↓ Peripheral vascular resistance by vasodilation of both resistance and capacitance vessels (arterioles and veins respectively)	↓ Blood pressure ↓ Venous return to the heart ↓ Preload
Decreased arterial pressure →	Compensatory Na ⁺ and water retention by kidney
Blocking contraction of the capacitance vessels enhances venous pooling →	↓ Ability to maintain BP in upright position → Orthostatic hypotension (syncope) Especially prominent with first dose or dose increase — “first dose effect” (may persist in some patients)

Practice points:

Alpha blockers should be taken with a diuretic to counteract sodium and water retention.
Check standing and recumbent blood pressure to check for persistent orthostatic hypotension
The first dose should be small and taken at bedtime to reduce the risk of falling.

Therapeutic uses of alpha blockers

Selective alpha-1 blockers

Prazosin

Terazosin

Doxazosin

- Hypertension (mild-moderate) refractory to first-line agents add-on to diuretic, BB, others
less protective in reducing CV morbidity/mortality
- Urinary retention due to benign prostatic hyperplasia
blocking α_1 receptors relaxes smooth muscle of the bladder neck and prostate, which improves urine flow

Nonselective alpha blockers

Phenoxybenzamine
(irreversible blocker)

- Pheochromocytoma to counteract hypertension / sweating due to excessive catecholamine release.
- Urinary outlet obstruction

Phentolamine
(competitive antagonist)

- Management of extravasation of vasopressors
- Reversal of local anesthesia containing vasoconstrictor (dental use)
- Diagnosis of pheochromocytoma (*obsolete*)

Alpha blockers adverse effects (not a complete list)

EFFECT	MECHANISM
Tachycardia by nonselective alpha blockers	α_2 AR block $\rightarrow$ NE release $\rightarrow$ stimulate cardiac β receptors $\rightarrow$ reflex tachycardia
➤ Selective alpha-1 blockers cause little tachycardia. Norepinephrine (NE) released from nerve endings acts on presynaptic α_2 ARs, which inhibits its own release.	
Miosis	Blocking ocular α_1 ARs blocks the mydriatic effect of sympathetic stimulation $\rightarrow$ unopposed muscarinic contraction of iris sphincter muscle $\rightarrow$ miosis
Nasal stuffiness	Blocking α AR-mediated constriction of nasal mucosa vasculature $\rightarrow$ nasal congestion
Intraoperative floppy iris syndrome	The cause has not been established.
Drug interactions	Other drugs that affect blood pressure

CENTRALLY ACTING SYMPATHOLYTICS

Alpha-2 Adrenergic Receptor *Agonists*

- *Clonidine

- *Methyldopa

Rarely use: Guanabenz and Guanfacine

(They have the same mechanism of action as clonidine.)

Clonidine / Methyldopa: Pharmacokinetic Properties

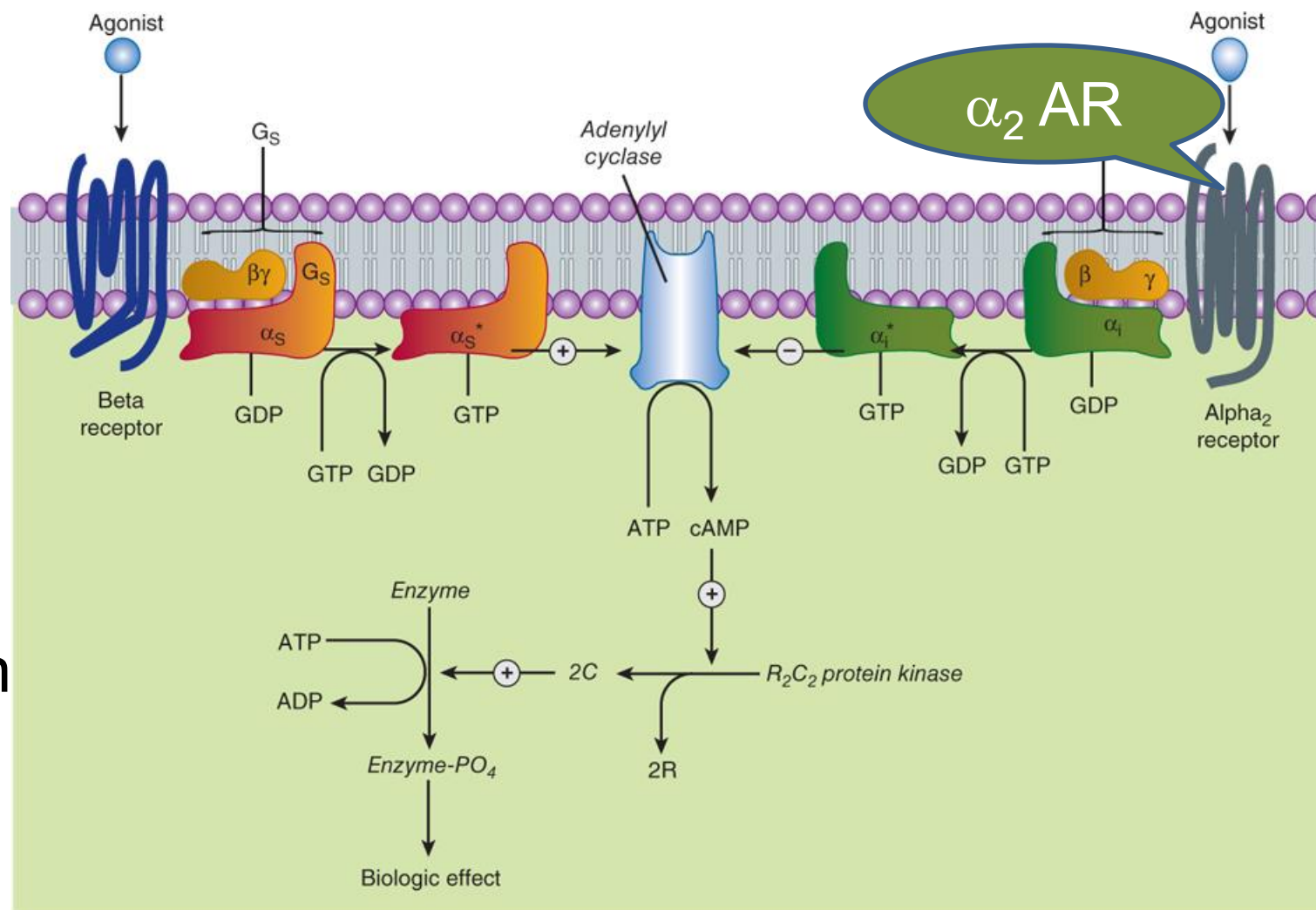
	CLONIDINE	METHYLDOPA
A	70-80% bioavailability	~42% bioavailability
D	Low plasma protein binding	Low plasma protein binding
M	Hepatic (inactive metabolites) enterohepatic circulation	Substrate of COMT Metabolism in gut and liver
E	Renal (~50% unchanged drug)	Renal (unchanged drug and metabolites)
t _{1/2}	12-16 hours	1.5 – 2 hours
Onset	0.5 – 1 hour	4-6 hours (oral and I.V.)
Duration	6-10 hours (dosing 2x daily)	24-48 h (dosing 2-3x daily)
Values refer to immediate release oral forms, adults, normal renal function Clonidine dosage forms: Oral tablets, extended release capsule, transdermal patch, solution for epidural injection		

α_2 -ARs are $G_{i/o}$ PCRs that inhibit adenylyl cyclase, $\downarrow$ cAMP, and $\downarrow$ PKA activity.

- They are likely also coupled to other signaling pathways

α_2 ARs are located on

- presynaptic,
- postsynaptic, and
- non-junctional sites in many tissues.



Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition Copyright © McGraw-Hill Education. All rights reserved.

$\alpha_{2A/2C}$ ARs inhibit neurotransmitter release (prejunctional autoreceptors and heteroreceptors).

Postsynaptic α_{2B} ARs induce vasoconstriction of arterioles and venules.

Baroreceptor controlled is retained.

Activation and inhibition of adenylyl cyclase by agonists that bind to catecholamine receptors. Binding to β adrenoceptors stimulates adenylyl cyclase by activating the stimulatory G protein, G_s , which leads to the dissociation of its α subunit charged with GTP. This activated α_s subunit directly activates adenylyl cyclase, resulting in an increased rate of synthesis of cAMP. Alpha₂-adrenoceptor ligands inhibit adenylyl cyclase by causing dissociation of the inhibitory G protein, G_i , into its subunits—ie, an activated α_i subunit charged with GTP and a β - γ unit. The mechanism by which these subunits inhibit adenylyl cyclase is uncertain. cAMP binds to the regulatory subunit (R) of cAMP-dependent protein kinase, leading to the liberation of active catalytic subunits (C) that phosphorylate specific protein substrates and modify their activity. These catalytic units also phosphorylate the cAMP response element binding protein (CREB), which modifies gene expression. See text for other actions of β and α_2 adrenoceptors.

Alpha-2 agonists reduce sympathetic outflow from CNS.

Agonists may:

1. activate postsynaptic α_2 ARs in the brainstem, inhibiting the activity of those neurons,
and / or
2. activate presynaptic α_2 reducing norepinephrine release,
and possibly
3. (Clonidine binding to imidazoline receptors may also contribute.)

activation of α_2 adrenergic receptors in the brainstem



↓ sympathetic and
↑ parasympathetic tone



lowers blood pressure
(↓HR and myocardial contractility),
causes bradycardia, and
decreases circulating
catecholamine levels

Alpha-2 agonists lower BP by decreasing heart rate, relaxing capacitance vessels and reducing peripheral vascular resistance.

Sensitivity to baroreceptor regulation is maintained.

Renal blood flow and GFR are essentially unchanged.

Clonidine

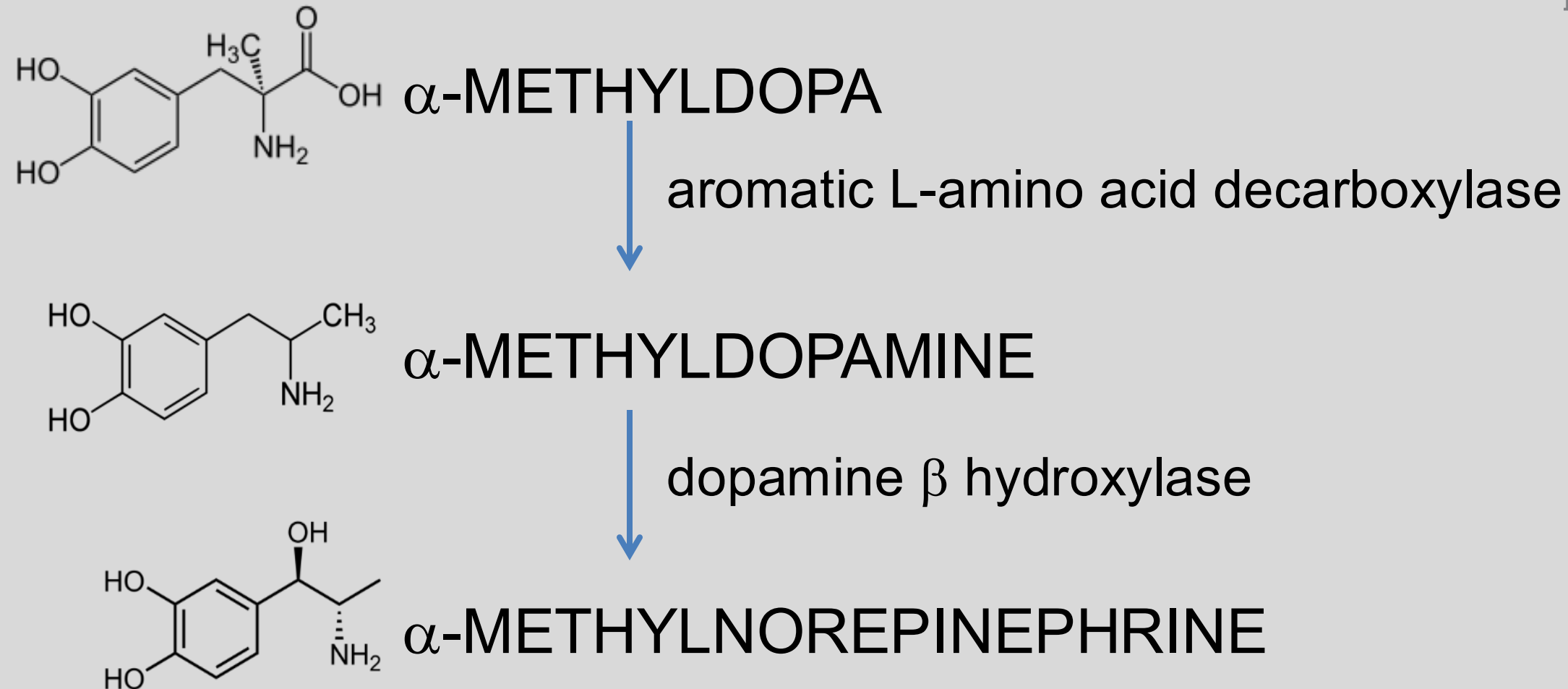
- ☐ Directly activates α_2 ARs in brainstem

Clonidine lowers heart rate and cardiac output more than methyldopa does, suggesting the sites of action may not be identical.

Methyldopa

- ☐ Taken up into presynaptic vesicles
- ☐ Converted to active methylnorepinephrine
- ☐ Replaces NE in secretory vesicle
- ☐ Stimulates α_2 ARs in the brainstem

Methyldopa is converted to active methylnorepinephrine



α MeNE is released instead of NE. It is called a “false transmitter”.

Alpha-2 Agonists Therapeutic uses

Clonidine

Refractory hypertension	Add-on after ACEI / ARB, diuretic, and CCB
Pain management	Epidural clonidine prevents pain signal transmission
ADHD	Alternative to SNRI therapy; the mechanism is unknown
Glaucoma	Ophthalmic drops reduce formation of aqueous humor
Substance abuse opioids, alcohol, smoking	Management of catecholamine-induced symptoms
Menopause	Management of vasomotor symptoms (hot flashes, sweating)

Methyldopa

Management of hypertension in pregnancy



Alpha-2 Agonists: Common and potentially serious adverse effects (this is not a complete list)

Patients with cardiovascular disease, i.e., bradycardia, coronary insufficiency, recent MI and conduction disturbances

Severe bradycardia and sinus arrest can occur in patients with SA nodal dysfunction or AV nodal block.

Common	Sedation, drowsiness, mental depression, constipation
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Epidural clonidine	Hypotension, respiratory depression
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→ **Avoid abrupt discontinuation of alpha-2 agonist. | Gradually taper clonidine.**
Sudden discontinuation → catecholamine surge → rebound hypertension
headache, apprehension, tremors, abdominal pain, sweating, and tachycardia

→ **Withdraw the beta blocker several days before clonidine is withdrawn.**
Removal of clonidine first would allow unopposed vasoconstriction.

DIRECT VASODILATORS

Oral

* Hydralazine

* Minoxidil

Parenteral

* Hydralazine

* Nitroprusside

* Fenoldopam

Diazoxide is no longer used for hypertension.

Pharmacokinetic Properties

	A	D	M	E	t _{1/2} ; onset; duration
Hydralazine	I.V. Oral B: 25% (variable)	85-90% protein binding	Extensive 1 st -pass effect Acetylation (genetic variability)	Urine, Metabolites	t _{1/2} : 2-8 h (oral) Onset: 20-30 min Duration: 8 h (oral) Duration: 1-4 h (I.V.)
Minoxidil	Oral B: 90%	not reported	Hepatic glucuronidation	Urine, Metabolites	t _{1/2} : ~4 hours Onset: 30 min Duration: ~3 days
Nitroprusside	I.V.	See next slide.....			
Fenoldopam	I.V.	not reported	Hepatic methylation, glucuronidation, sulfation	Urine, Metabolites	t _{1/2} : ~5 minutes Onset: 10 min Duration: 1 hour

A: Absorption; B: bioavailability; D: distribution; M: metabolism; E: excretion; t_{1/2}: adults, normal renal function

Nitroprusside Pharmacokinetics

Nitroprusside is taken up
by red blood cells

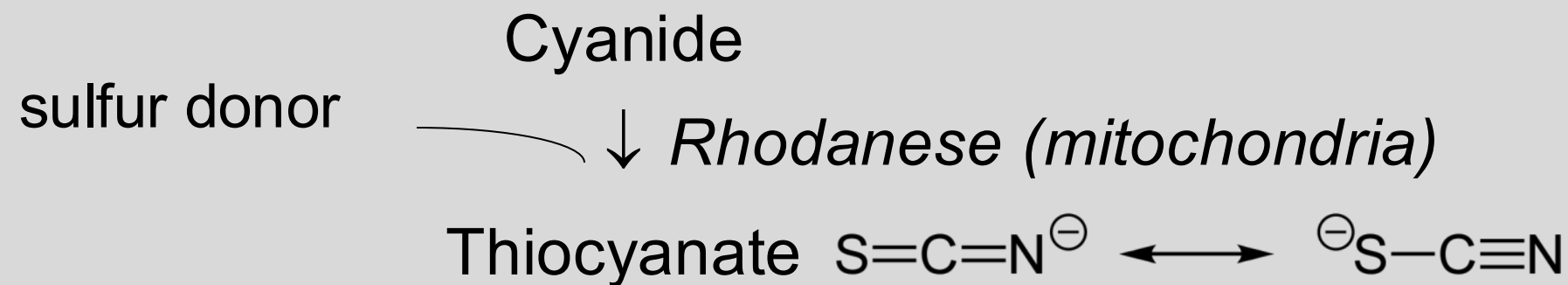


NO + cyanide (CN⁻)

Saturable process:

Large exposures of cyanide exhaust
sulfur donor supplies → ***cyanide
toxicity***

Cyanide detoxification (biological):



$t_{1/2}$ elimination:

Nitroprusside, circulatory: ~2 min

Thiocyanate, renal elimination: ~3 days
(may ↑2-3x in renal failure)

Hypotensive effect:

Onset: <2 min

Duration: 1-10 min

Hydralazine Pharmacogenomics: N-Acetyltransferase 2 (NAT2)

Slow acetylators...

may be more subject to adverse effects

⇒ *Clinical correlate:*

- Slow acetylators are at risk of developing antinuclear antibodies and lupus-like syndrome

Rapid acetylators...

have greater first-pass metabolism and lower blood levels

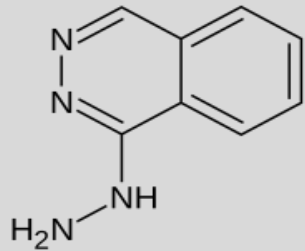
⇒ *Clinical correlate:*

- Rapid acetylators may be more prone to inadequate therapeutic response to standard drug doses → less benefit from standard doses

Vasodilators decrease BP

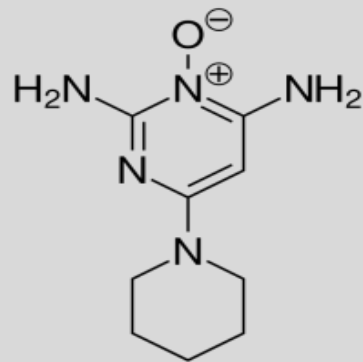
by decreasing systemic vascular resistance

Hydralazine



Relaxes vascular smooth muscle
(mechanism not defined)
Dilates arterioles but not veins

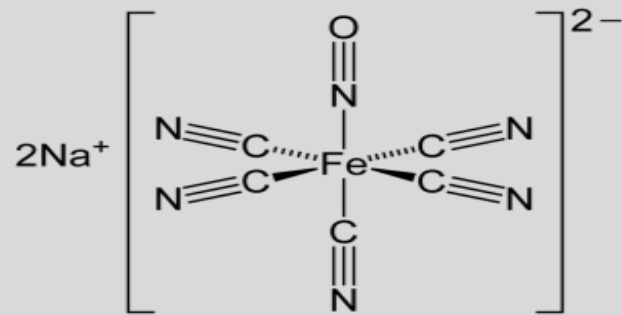
Minoxidil sulfate



Active metabolite mediates opening of K_{ATP} channels in
smooth muscle membranes of arterioles →
hyperpolarization → relaxation of smooth muscle
Dilates arterioles but not veins

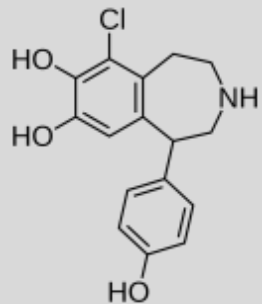
Vasodilators decrease BP by decreasing systemic vascular resistance

Nitroprusside



Activates guanylyl cyclase, either via release of nitric oxide or by direct stimulation of the enzyme → ↑ intracellular cGMP → vascular smooth muscle relaxation
Dilates arterial and venous vessels

Fenoldopam



Dopamine D1 agonist → relaxation of vascular smooth muscle + natriuresis / diuresis
Dilates peripheral arterioles

D₁R: G_sPCR → activates adenylyl cyclase → ↑ cAMP

Direct vasodilators (oral), hydralazine and minoxidil, for refractory hypertension

Relaxing the smooth muscle of arterioles

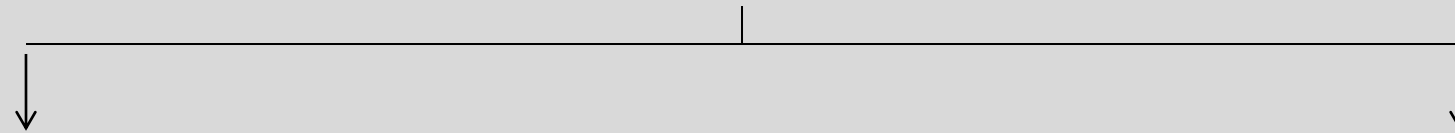


↓ SVR and ↓ MAP

(systemic vascular resistance) ↓ (mean arterial pressure)

compensatory responses

mediated by baroreceptors and the SNS, renin, ANG II, and aldosterone



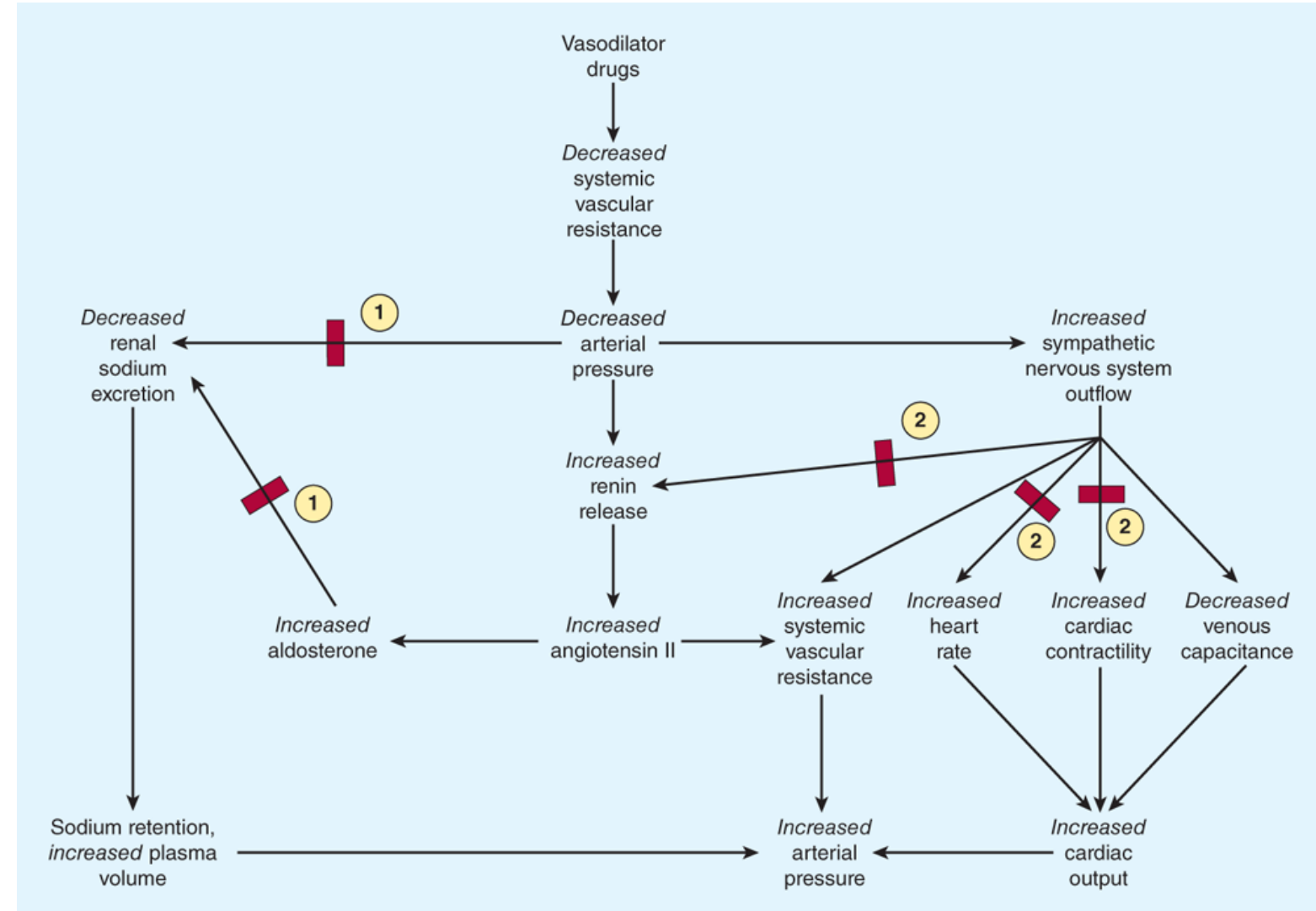
Vasodilators work best in combination with other antihypertensive drugs that oppose the compensatory cardiovascular responses.

Vasodilator therapy does not cause orthostatic hypotension or sexual dysfunction because sympathetic reflexes are intact.

Compensatory responses to vasodilators

Vasodilators are given with other agents to:

- increase efficacy and
 - counteract the compensatory responses:
 - salt and water retention, tachycardia –
- 1) diuretics
- 2) beta blockers



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Compensatory responses to vasodilators; basis for combination therapy with β blockers and diuretics. Effect blocked by diuretics. Effect blocked by β blockers.

Vasodilators Therapeutic Uses

Hydralazine	Moderate to severe hypertension Heart failure with reduced ejection fraction: hydralazine + nitrate such as isosorbide dinitrate (evidence of benefit in black individuals) Hypertensive emergency in pregnancy or postpartum
Minoxidil	Severe hypertension (last line); strong antihypertensive effects
Nitroprusside	Acute hypertension Acute decompensated heart failure Controlled hypotension during surgery
Fenoldopam	Severe hypertension (treatment up to 48 hours) May be used in patients with renal compromise and pediatric patients (short-term)



- Headache, nausea, anorexia, palpitations, sweating, flushing
- Reflex tachycardia – *may provoke angina or ischemic arrhythmia*
- Salt and water retention – *may lead to development of congestive heart failure*

Beta blocker and diuretic should be co-administered to counteract these compensatory effects.

Note: Because of preferential dilation of arterioles over veins, sympathetic reflexes are intact; postural hypotension and sexual dysfunction are not common problems.

Hydralazine	Lupus-like syndrome – slow acetylators – rash, arthralgia; reversible
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Minoxidil	Hypertrichosis – minoxidil topically for hair growth due to male pattern baldness
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Vasodilators “Unique” Adverse Effects

Hydralazine	Lupus-like syndrome, reversible – rash, arthralgia, myalgia, fever More likely with prolonged therapy and among slow acetylators	
Minoxidil	Hypertrichosis – Topically for hair growth due to male pattern baldness	
Nitroprusside	Excessive BP lowering Cyanide accumulation	<i>Antidotes to cyanide accumulation:</i> 1. Sod. nitrite f/b sod. thiosulfate → sulfur donor → cyanide metabolism 2. Hydroxocobalamin → cyanocobalamin (nontoxic vitamin B12)
Fenoldopam	↑ Intraocular pressure Avoid in glaucoma patients	



Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine*, 8th Edition: www.accessmedicine.com

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Localized acute cutaneous lupus erythematosus. Erythematous, slightly edematous, sharply demarcated erythema is seen on the malar areas in a “butterfly” distribution



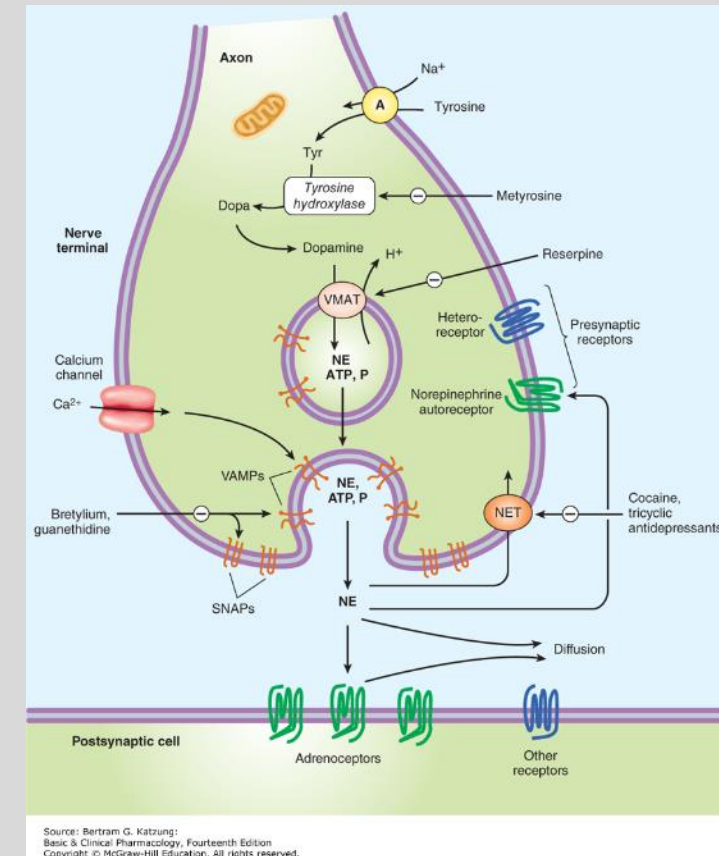
Source: Wolff K, Johnson RA, Saavedra AP: *Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology*, Seventh Edition: www.accessmedicine.com
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Hypertrichosis of face; excessive hair growth in nonandrogen-sensitive areas of the face in a female treated with cyclosporine

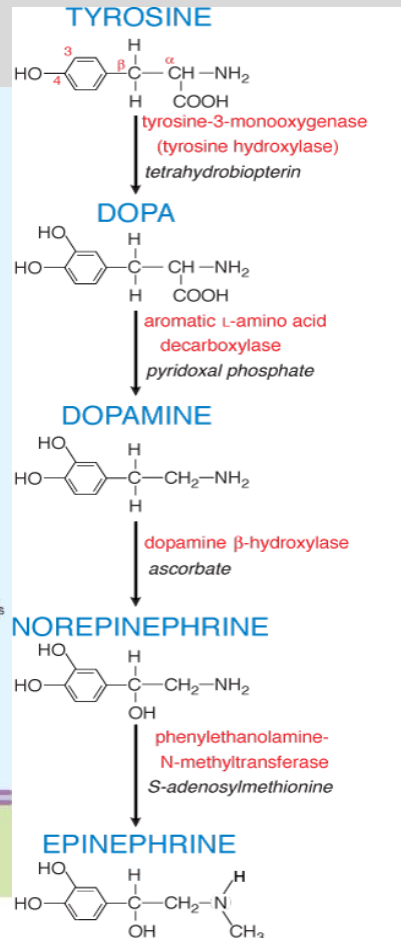
Drugs that cause catecholamine depletion in sympathetic nerve terminals are Reserpine, Guanethidine, Guanadrel, and Metyrosine. They are rarely used and will not be covered here.

Briefly, for your information:

- Reserpine is an irreversible VMAT2 inhibitor, which causes depletion of neurotransmitter centrally and peripherally. Adverse effects include orthostatic hypotension, sexual dysfunction, and sedation and mental depression.
- Guanethidine and guanadrel replace NE in storage vesicles, leading to NE depletion. Neither drug is available in the US.
- Metyrosine blocks tyrosine hydroxylase, the rate limiting step in the biosynthesis of catecholamines, which decreases the levels of endogenous catecholamines.



Source: Bertram G. Katzung: Basic & Clinical Pharmacology, Fourteenth Edition Copyright © McGraw-Hill Education. All rights reserved.



Source: Brunton LL, Chabner BA, Knollmann BC: Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition: www.accessmedicine.com Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

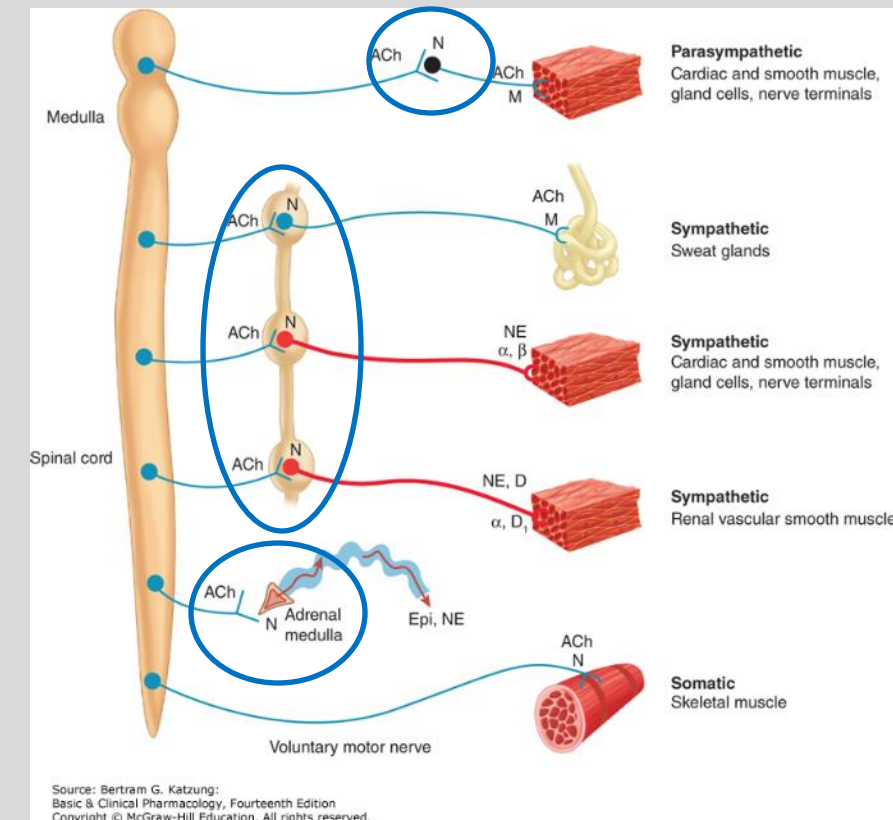
Likewise, Ganglion-blocking drugs mecamylamine (oral) and trimethaphan (I.V.) are no longer in use and will not be covered here.

Briefly for your information:

Ganglion blockers were among the first agents used in the treatment of hypertension. They are no longer available clinically because of intolerable toxicities related to their primary action – blocking ganglionic nicotinic receptors (circled on the figure). The drugs block decrease peripheral resistance and cardiac output by blocking all autonomic outflow.

Adverse effects:

- Sympathoplegia: excessive orthostatic hypotension
- Parasympathoplegia: constipation, urinary retention, precipitation of glaucoma, blurred vision / mydriasis, dry mouth
- Sexual dysfunction: and decreased libido, impotence (both erection and ejaculation)
- CNS: altered mental status, choreiform movements, convulsions, fatigue, orthostatic dizziness, paresthesia, sedation



Are you a matchmaker?

On the following slides, match the drugs to their mechanisms and effects.

How do the effects relate to their mechanisms?

Match the drug to the mechanism

Aliskiren	alpha-1 adrenergic receptor antagonist
Captopril	alpha-2 agonist
Clonidine	angiotensin AT1 antagonist
Fenoldopam	angiotensin converting enzyme inhibitor
Hydralazine	beta adrenergic receptor antagonist
Losartan	direct arteriolar dilator
Mecamylamine	direct renin inhibitor
Minoxidil	dopamine D1 receptor agonist
Nifedipine	ganglion blocker
Nitroprusside	guanylyl cyclase activation
Prazosin	K _{ATP} channel blocker
Propranolol	L-type Ca ²⁺ channel blocker with cardiac effects
Reserpine	L-type Ca ²⁺ channel blocker, vasodilatory effects
Verapamil	VMAT inhibitor

Match the drug to the mechanism (answers)

Aliskiren	Direct renin inhibitor
Captopril	ACE inhibitor
Clonidine	Alpha-2 agonist
Fenoldopam	Dopamine D1 receptor agonist
Hydralazine	Direct arteriolar dilator
Losartan	Angiotensin AT1 receptor blocker
Mecamylamine	Ganglion blocker
Minoxidil	K _{ATP} channel blocker
Nifedipine	L-type Ca ²⁺ channel blocker, vasodilatory effects primarily
Nitroprusside	guanylyl cyclase activation
Prazosin	Alpha-1 adrenergic receptor antagonist (selective)
Propranolol	Beta adrenergic receptor blocker (nonselective)
Reserpine	VMAT inhibitor
Verapamil	L-type Ca ²⁺ channel blocker with cardiac effects

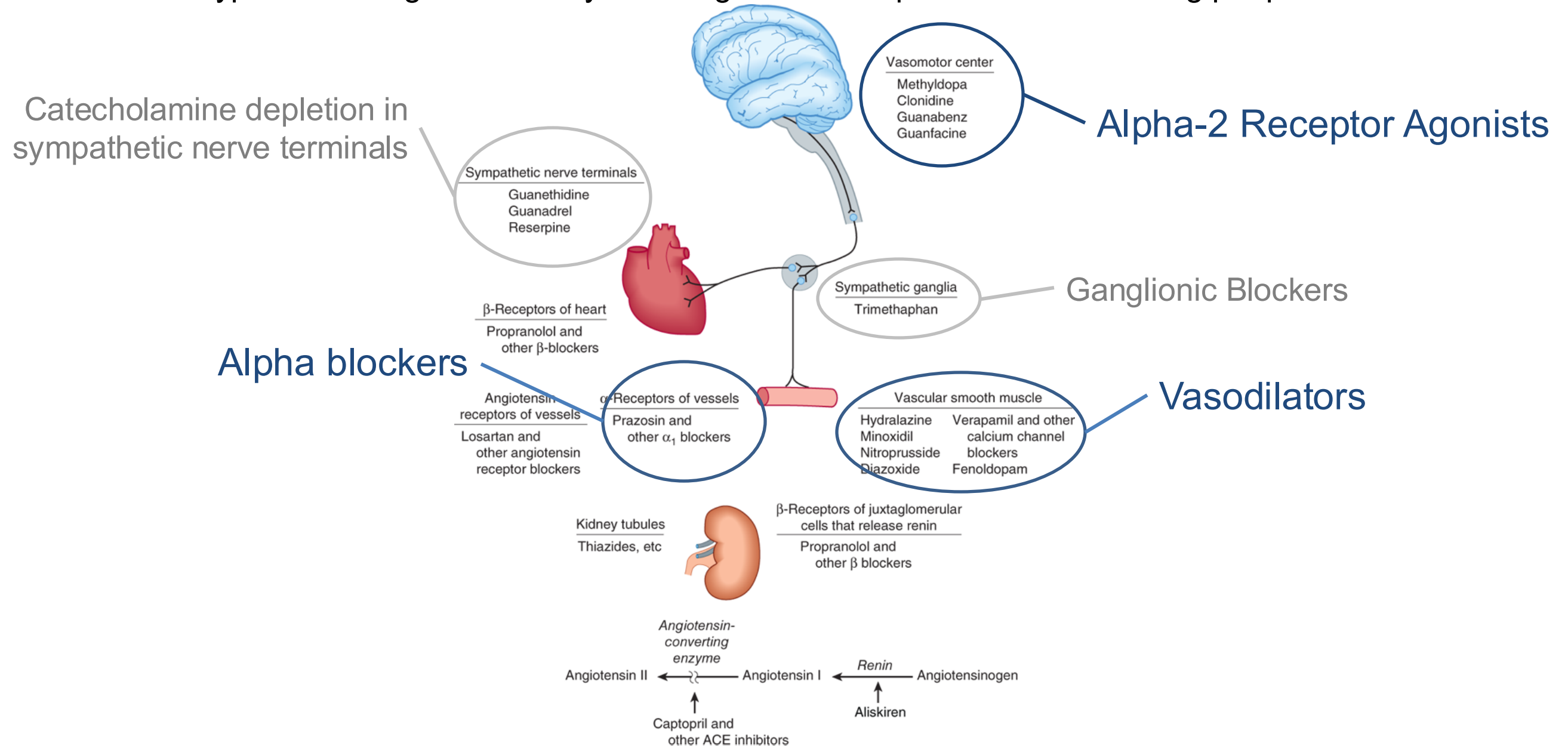
Match the drug to the adverse effect(s)

Aliskiren	Acute kidney injury	Hemolytic anemia
Captopril	Angioedema	Hyperkalemia
Clonidine	Bradycardia	Hypertrichosis
Hydralazine	Bronchospasm	Hypotension, excessive
Fenoldopam	Constipation	Lupus-like syndrome
Losartan	Cough	Orthostatic hypotension
Mecamylamine	Cyanide toxicity	Reflex tachycardia
Minoxidil	Depression (severe)	Sedation
Nifedipine	Dry mouth	Sexual dysfunction
Nitroprusside	Edema	Teratogenic
Prazosin	GERD	
Propranolol		
Reserpine		
Verapamil		

Summary of Alpha Blockers Centrally Acting Sympatholytics Vasodilators

Sites of action of the major classes of antihypertensive drugs.

Most antihypertensive agents work by reducing cardiac output and/or decreasing peripheral resistance.



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- The α_1 -adrenergic receptor competitive antagonists (alpha blockers) used in the management of hypertension are prazosin, terazosin, and doxazosin. Terazosin and doxazosin have longer $t_{1/2}$ than prazosin and may be taken once daily. They decrease peripheral vascular resistance by causing relaxation of arterial and venous smooth muscle, which reduces preload. The drugs cause only minimal changes to cardiac output, renal blood flow, and glomerular filtration.
- Orthostatic hypotension and reflex tachycardia are common with initiation of therapy – first dose effect. Salt and water retention occur, which is mitigated when the alpha blocker is given with a diuretic.
- Alpha-1 blockers are add-on drugs for the treatment of refractory hypertension. They are not recommended initial treatment due to weak clinical trial data.
- Alpha-1 blockers, terazosin and doxazosin, may also be useful in managing the urinary symptoms of benign prostatic hyperplasia.
- Nonselective alpha blockers are useful in the perioperative management of pheochromocytoma. They cause orthostatic hypotension and tachycardia.

- Alpha-2 agonists inhibit sympathetic vasomotor centers, which reduces sympathetic outflow from the CNS (central sympatholytic effect). Reduction in cardiac output (decreased heart rate) and peripheral vascular resistance and relaxation of capacitance vessels results in lowering of blood pressure. It does not appreciably decrease renal blood flow or glomerular filtration rate.
- Clonidine is administered as oral tablets and long-acting patch for the management of hypertension. Epidural clonidine is used for pain management. Clonidine is also used for managing sympathetic symptoms associated with substance withdrawal and may relieve vasomotor symptoms associated with menopause.
- Abrupt withdrawal can cause rebound hypertension. Beta blockers should be tapered and withdrawn before tapering and withdrawal of clonidine
- Common adverse effects include sedation, dry mouth, bradycardia, and constipation.
- Methyldopa is converted in sympathetic neurons to α -methylnorepinephrine, which displaces norepinephrine in storage vesicles. It activates central α 2 receptors. It is safe in pregnancy. Side effects include sedation, bradycardia, and constipation.

- Direct-acting vasodilators hydralazine and minoxidil relax smooth muscle of the arteries and arterioles, which decreases peripheral vascular resistance, lowering blood pressure. They are used for the management of refractory hypertension. Minoxidil is a powerful antihypertensive agent.
- They cause reflex tachycardia, which could prompt angina, myocardial infarction, and heart failure. They cause salt and water retention. Concurrent beta blocker and diuretic are recommended to counteract these effects.
- Hydralazine with a nitrate is also used in the management of heart failure. Hydralazine is also used in the management of hypertension in pregnancy and postpartum.
- Adverse effects of hydralazine include headache, palpitations, and sweating. Reversible lupus-like syndrome can occur with high dose and poor acetylators (hydralazine is metabolized by NAT2).
- Minoxidil can cause hypertrichosis. Topical minoxidil is used to grow hair.

- Nitroprusside is a powerful intravenous vasodilator for the management of hypertensive emergencies. It causes the release of nitric oxide, dilating both venous and arterial blood vessels. Nitroprusside reduces peripheral vascular resistance and venous return. Excessive hypotension can result in death. Nitroprussides blood pressure lowering effects rapidly disappear when the infusion is stopped.
- The most serious toxicity is accumulation of cyanide, which can occur with prolonged infusion or defect in cyanide metabolism. Sodium thiosulfate is a sulfur donor that facilitates cyanide metabolism. Hydroxocobalamin combines with cyanide to form nontoxic cyanocobalamin.
- Fenoldopam is an intravenous dopamine D1 receptor agonist that induces dilation of peripheral arteries and natriuresis. It has a short duration of action (half-life ~10 minutes). It is used for hypertensive emergencies.
- Fenoldopam causes reflex tachycardia, headache, and flushing, and increases intraocular pressure (caution in patients with glaucoma).



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 - Chapter 8: Cholinoceptor Blocking Drugs
 - Chapter 9: Adrenoceptor Agonists & Sympathomimetic Drugs
 - Chapter 10: Adrenoceptor Antagonist Drugs Chapter 11: Antihypertensive Agents
- Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics, 14e, 2023
 - Chapter 14. Adrenergic Agonists and Antagonists
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